

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2023 - 2024)

JEE(Advanced)**PRE-SCORE****08-10-2023****JEE(Main+Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)**

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 180

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY**GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

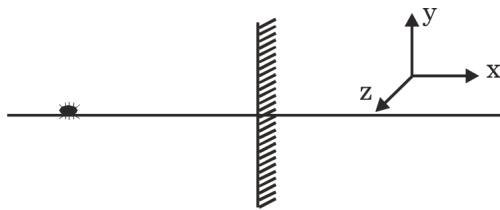
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

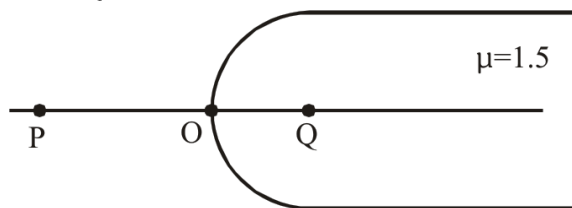
Negative Marks : -1 In all other cases

1. A bug is moving with velocity $(2\hat{i} - 3\hat{j} + 4\hat{k})$ m/s in front of plane mirror which is also moving with velocity $(2\hat{i} + \hat{j} + \hat{k})$ m/s. Plane mirror always remain parallel to y-z plane. Velocity of image of bug is :-

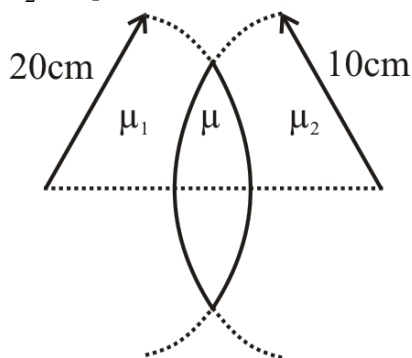


- (A) $2\hat{i} - 3\hat{j} + 4\hat{k}$
 (B) $-2\hat{i} - 3\hat{j} + 4\hat{k}$
 (C) $-2\hat{i} + 3\hat{j} - 4\hat{k}$
 (D) None
2. Ram is looking at his face in a mirror kept 10 cm away and he finds that his image is erect and magnified ($m = 1.8$). If he holds the mirror 50 cm away :
- (A) He cannot see the image because reflected rays falling on his eyes are converging
 (B) He sees a magnified and erect image
 (C) He sees a diminished and inverted image
 (D) He sees a magnified and inverted image.

3. One end of a glass rod of refractive index $n = 1.5$ is a spherical surface of radius of curvature R . The centre of the spherical surface lies inside the glass. A point object placed in air on the axis of the rod at the point P has its real image inside glass at the point Q (see fig.). A line joining the points P and Q cuts the surface at O such that $OP = 2OQ$. The distance PO is :-



- (A) $8R$ (B) $7R$
(C) $2R$ (D) None of these
4. Radii of curvature of surfaces of a biconvex lens are 10 cm and 20 cm and refractive index of its material is $\mu = 1.5$. If the refractive indices of the medium on the two sides of the lens are μ_1 and μ_2 , then the parallel rays incident on the lens from left will focus at a distance of f_2 (distance being measured from the optical centre of the lens) where f_2 is equal to :-



- (A) $\frac{20\mu_1}{4.5 - (2\mu_1 + \mu_2)}$ (B) $\frac{20\mu_2}{4.5 - (\mu_1 + 2\mu_2)}$
(C) $\frac{20\mu_2}{4.5 - (2\mu_1 + \mu_2)}$ (D) $\frac{20\mu_1}{4.5 - (\mu_1 + 2\mu_2)}$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

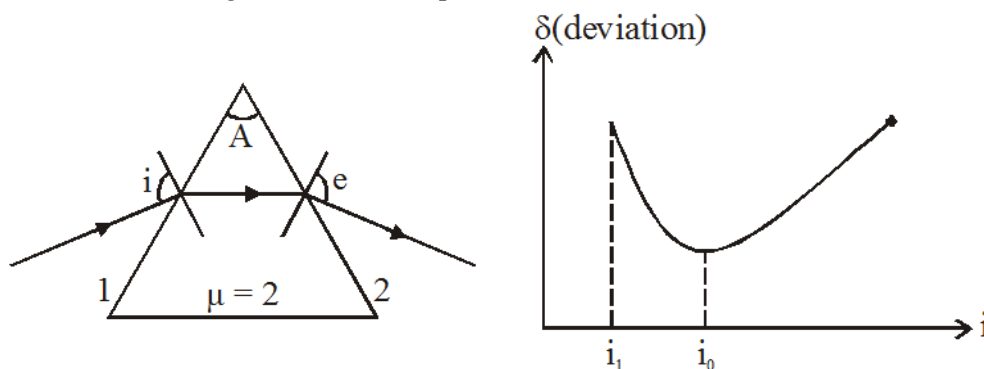
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

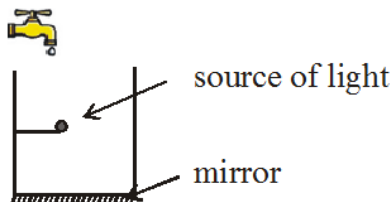
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. A monochromatic light is incident on a prism as shown. Deviation suffered by ray varies with angle of incidence as shown. Assuming refraction takes place at 1 & 2. Choose the correct alternative(s).



- (A) i_1 can be equal to zero if $A = 30^\circ$
- (B) If angle of incidence is increased by 2° , then angle of emergence may decrease by less than 2° .
- (C) If angle of incidence is increased by 2° , then angle of emergence may decrease by more than 2° .
- (D) If angle of incidence is increased by 2° , then angle of emergence may increase by 2° .

6. A small source of light is mounted inside a cylindrical container of height h . The bottom of the container is covered with a mirror. Initially, the container is empty. Then a clear liquid with the index of refraction n is slowly poured into the container. The level of liquid H rises steadily, reaching the top of the container in time T . Let h_1 be the distance of source of the light from the bottom of the container. Consider paraxial ray approximation. The observer is vertically above the source in air and he is observing image after reflection.



Consider two cases

- (1) $H \leq h_1$ (2) $h_1 \leq H \leq h$

- (A) The speed of the image of the source in case (1) during this process is $\frac{2h}{T} \left(1 - \frac{1}{n}\right)$.
- (B) The speed of the image of the source in case (2) during this process is $\frac{h}{T} \left(1 - \frac{1}{n}\right)$.
- (C) The speed of the image of the source in case (1) during this process is $\frac{h}{2T} \left(1 - \frac{1}{n}\right)$.
- (D) The speed of the image of the source in case (2) during this process is $\frac{h}{2T} \left(\frac{1}{n}\right)$.
7. Consider an equilateral prism of refractive index μ . It is found that when light is incident at an angle 60° , angle of emergence is 'e', while when light is incident at an angle 50° , it undergoes TIR on the 2nd refracting surface. Choose the **CORRECT** options :-
- (A) A ray incident at an angle 65° will surely emerge from other surface.
- (B) A ray incident at an angle of 40° will surely suffer TIR on other surface.
- (C) A ray incident at an angle of 50.7° , will surely suffers TIR on other surface.
- (D) A ray incident at an angle 59° , may suffer TIR on the other surface.

8. Two mirrors are inclined at an angle of 30° one of the ray is incident parallel to the one of the mirror. Then select the correct statement(s)-
- (A) Ray will undergo five reflections from the system
 - (B) Total deviation from the system will 120°
 - (C) Ray will retrace its path finally
 - (D) Total deviation suffered by the ray will be 90°
9. A point source of light is located on the optical axis of a thin converging lens of aperture diameter 1 cm with focal length 10 cm. After the refraction from the lens
- (A) The maximum angle of diverging beam possible is 0.1 radians.
 - (B) The maximum angle of a converging beam possible is 0.1 radians.
 - (C) The minimum possible distance between the real image and the object is 40 cm.
 - (D) The maximum possible angle of deviation for any ray is 0.05 radians.
10. An observer holds in front of himself a thin equi convex converging lens; a is the numerical value of the radius of curvature of each face. He sees two clear and distinct images of one of his eyes, one erect and the other inverted. The refractive index of lens is 1.50. Least distance of distinct vision is 25 cm. One image is due to reflection from front surface & other is due to reflection from rear surface.
- (A) The observer must be at a distance more than $0.5a$ from the lens.
 - (B) The distance between observer and his inverted image must be more than 25 cm.
 - (C) If the observer comes at a distance more than $0.25a$ but less than $0.5a$ from the lens, he sees two erect images.
 - (D) If the observer comes at a distance less than $0.25a$ from the lens, he can see two erect images.

SECTION-II (i) : (Maximum Marks: 12)

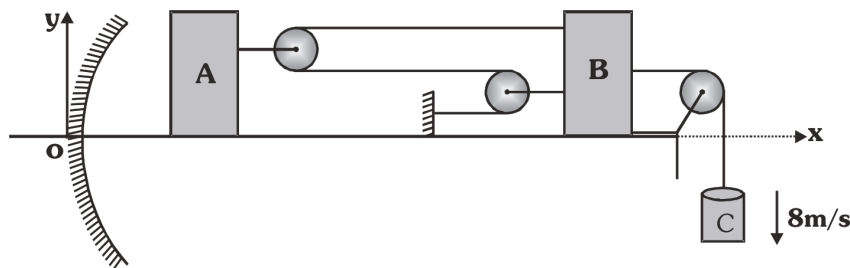
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

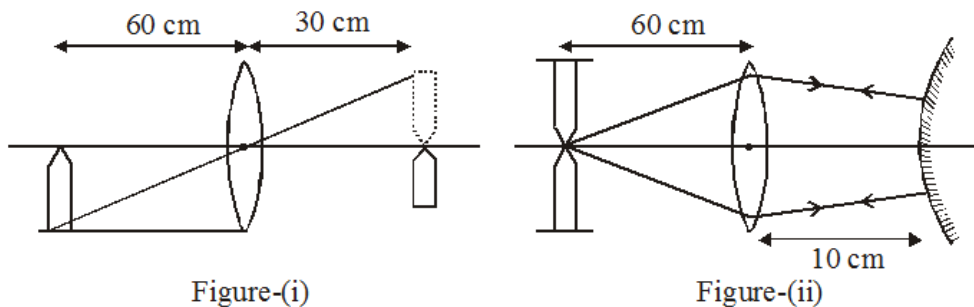
For the given system of blocks and a concave mirror answer the following. (Assuming A, B & C are very small). Focal length of mirror is 20 cm height of A & B are same.



1. If A is at $x = 10$ cm then image of B and image of A are of same size. What is the distance of B (in cm) from mirror ?
2. Velocity of image of A when it is at $x = 30$ cm is _____ m/s

Paragraph for Questions 3 and 4

An experiment is performed to calculate the focal length of a convex mirror using convex lens. The setup consists of two pins, optical bench, one convex lens and one convex mirror. When only convex lens is placed at a distance 60 cm from pin (object), its image is formed at a distance 30 cm from the lens as shown in figure-(i). Now convex mirror is also kept right to the lens without changing distance between pin (object) and lens. The mirror is kept at a distance 10 cm from lens such that position of image of object pin and position of object pin coincides as shown in figure-(ii)



3. Focal length of convex lens used (in cm) is :
4. Focal length of convex mirror (in cm) is :

Paragraph for Questions 5 and 6

Assume that each grade of glass is characterized by a refractive index n_0 at a wavelength $\lambda_0 = 0.55 \mu\text{m}$ and a coefficient dispersion β , which shows how much the refractive index of a given type of glass depends on wavelength λ . $n_\lambda = n_0 + \beta(\lambda - \lambda_0)$. There are two grades of clear glasses with characteristics $\beta_1 \neq \beta_2$. We want to make lenses for glasses that lack chromatic aberration, i.e. combined optical power is not dependent on the wavelength. For this we need to combine two or more lens. Assume that we have combined a bi-convex lens with a convexo-concave lens. Radii of bi-convex lens is R_1 and R_2 and that for convexo-concave lens is R_2 and R_0 . The lens exactly fit into each other. $\beta_1 = 0.05 \text{ nm}^{-1}$, $\beta_2 = 0.06 \text{ nm}^{-1}$, $R_0 = 20 \text{ cm}$, $R_1 = 10 \text{ cm}$, $n_{01} = n_{02} = 1.5$

5. What is the value of R_2 (in cm)?
6. If $\beta_1 = 0.05 \text{ nm}^{-1}$, $\beta_2 = 0.06 \text{ nm}^{-1}$, $R_0 = 20 \text{ cm}$, $R_1 = 10 \text{ cm}$, $n_{01} = 1.5$ and $n_{02} = 1.6$ $\lambda_0 = 500 \text{ nm}$. What is the value of R_2 (in cm) for which the ray of wavelength 520 nm passes through the combination without deviation?

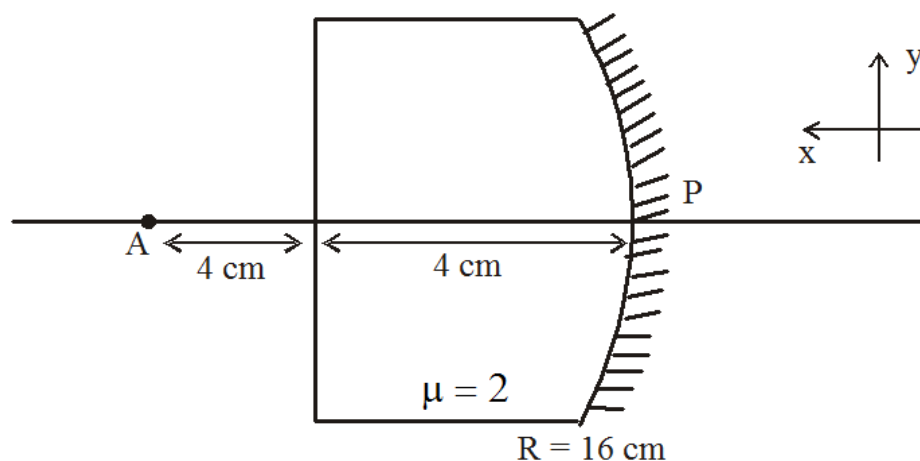
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

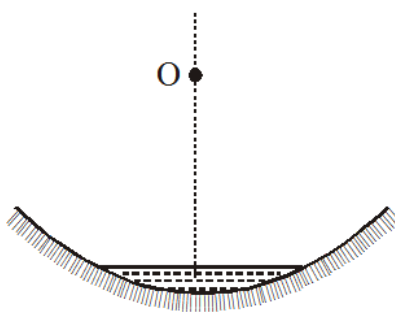
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A curved thick glass surface is silvered at curved face & not silvered on plane surface. Object is placed at A as shown in figure. Considering P (pole of the silvered surface) as origin. If x - co-ordinate of final image is 'n' cm then find n.



8. A flint glass convex lens of focal length 16 cm is placed in contact with crown glass lens. The refractive indices of crown glass for violet and red colour are $\mu_v = 1.525$, $\mu_r = 1.515$ and for flint glass $\mu_v = 1.655$ and $\mu_r = 1.645$. This forms an achromatic combination with the flint glass lens. What is the focal length of the combination (in cm) ?
9. The concave spherical mirror of radius of $R = 9$ cm forms the image of a source which coincides with a source. When a little liquid is poured in the mirror, a second image is formed between the source and the mirror at a distance $\ell = 4$ cm from the source. Find the refractive index of the liquid. Write '5μ' in OMR sheet.



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. **Statement-1** : $\Delta_f H$ of H is 1312 kJ/mol.

Statement-2 : Reactivity of hydrogen is very low as compared to halogens.

Statement-3 : Deuterium exist mostly in terms of HD.

Statement-4 : Tritium is radioactive and emits high energy β particles.

Statement-5 : H-H bond dissociation enthalpy is lowest for a single bond.

Consider above statements and select correct scheme (T = True, F = False)

(A) T T F T F

(B) T T T T F

(C) F T T T T

(D) T T T F F

2. Select correct statement :

(A) α -nitroso- β -naphthol is chelating ligand.

(B) Cupron is ligand with only N-donor sites.

(C) D₂O is used in production of cephalosporin.

(D) Certain plants like Pinus, Juniparus, Pyrus can metabolise sulphur oxides.

3. Select the incorrect statement about ice :

(A) At atmospheric pressure ice crystallises in the Hexagonal form.

(B) At very low temperatures ice condenses to Cubic form.

(C) Ice formed by heavy water sinks in normal water.

(D) In ice each oxygen atom is surrounded tetrahedrally by four other oxygen at two type of distance 95.7 pm and 276 pm.

4.

1.	Ni(CO)_4	(a)	A complex ion having a magnetic moment of 2.83 Bohr magnetons
2.	$[\text{Ni(CN)}_4]^{2-}$	(b)	34
3.	$[\text{Ni(NH}_3)_6]^{2+}$	(c)	Ziese's salt
4.	$[\text{Zn(NH}_3)_4]^{2+}$	(d)	cis-trans Isomerism
5.	$[\text{Pd(NH}_3)_2\text{Cl}_2]$	(e)	54
6.	$\text{K}^+[\text{Pt(C}_2\text{H}_4)\text{Cl}_3]^-$	(f)	A low-spin (spin-paired) complex
7.	$[\text{Co(NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}$	(g)	A complex ion having a magnetic moment of 1.73 Bohr magnetons
8.	EAN for $[\text{Ni(CN)}_4]^{2-}$	(h)	Linear complex
9.	EAN for $[\text{Pd(NH}_3)_6]^{4+}$	(i)	Shows geometrical isomerism
10.	$[\text{Co(NH}_3)_6]^{3+}$	(j)	A diamagnetic species
11.	CoF_6^{3-}	(k)	Square planar
12.	$[\text{Mn(CN)}_6]^{4-}$	(l)	A high-spin (spin-free) complex
13.	$[\text{V(H}_2\text{O)}_6]^{3+}$	(m)	Tetrahedral
14.	$[\text{Ag(NH}_3)_2]\text{Cl}$	(n)	A paramagnetic species

Consider above given list.

Select option in which only correct matches given :

(A) 1-m, 2-k, 3-h, 8-b, 9-e, 10-d

(B) 3-h, 4-j, 5-i, 11-l, 12-g, 13-j

(C) 13-a, 14-h, 9-e, 10-f, 3-j, 4-j

(D) 7-d, 8-b, 2-k, 3-n, 12-g, 6-c

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Select the correct statement(s) about actinoids :

- (A) Actinoids gives mixture of oxides and hydrides on boiling with water.
- (B) Most of actinoids are slightly affected by nitric acid but react with HCl.
- (C) Shows violent reaction with NaOH.
- (D) Lanthanoid contraction is greater from element to element than actinoid contraction.

6. Symmetrical Di- μ -hydroxytetrakis(ethylenediamine)dnicobalt(III) chloride (P) complex reacts with aq. HCl to give a product $\text{Co(en)}_2\text{Cl}_3$ (Q) which is resolvable with optical isomers. When it is kept in acidic medium the compound changes color and give isomer (R) which is not optically resolvable. Select correct statement(s) ?

- (A) If one 'en' is replaced by two ' NH_3 ' molecules then complex (R) become optically resolvable.
- (B) If one 'en' is replaced by two ' NH_3 ' molecules and second 'en' is replaced by two H_2O molecules. Complex 'R' still optically not resolvable.
- (C) If one 'en' is replaced by two ' NH_3 ' molecules in complex 'Q', complex is still optically resolvable.
- (D) If one 'en' is replaced by two ' NH_3 ' then second 'en' is replaced by two ' H_2O ' in complex 'Q' become optically not resolvable.

7. A compound (X) is made by mixing Cobalt(III) nitrite and potassium nitrite solution in ratio of 1 : 3. The aqueous solution of the compound showed 4 particles per molecule where as molar conductivity reveals presence of six electrical charges.
Select correct statement about complex (X) :
- (A) Complex 'X' is low spin complex.
(B) Complex 'X' can show maximum '10' isomeric forms.
(C) Complex 'X' can show '3' optically active isomers.
(D) Complex 'X' is inner orbital complex.
8. The Co(III) aqua complex slowly undergoes redox decomposition to give the Co(II) aqua complex and oxygen, but in presence of 'en', Co(II) is slowly oxidised to $[\text{Co(en)}_3]^{3+}$ by aerial oxygen.
Select correct statement(s) :
- (A) After redox decomposition magnetic moment decreases.
(B) After arial oxidation magnetic moment increased.
(C) After arial oxidation crystal field splitting energy increased.
(D) After redox decomposition crystal field splitting energy decreased.
9. Select the correct statement(s) ?
- (A) $[\text{Co(gly)}_2(\text{OH}_2)_2]^+$ can produce 5 geometrical isomers.
(B) $[\text{Tris(tetramine-}\mu\text{-dihydroxidocobalt(III))}]$ cobalt(III) ion shows optical activity.
(C) $[\text{PtCl}_2(\text{NH}_3)_2(\text{NO}_2)_2]$ can have maximum 6 isomers.
(D) For $[\text{Co(Br)(en)(NH}_3)_2(\text{OH}_2)]^{2+}$ ion total geometrical isomers is equal to the number of optically active isomers.
10. Select the incorrect statement(s) :
- (A) $[\text{Cd}(\text{CH}_3\text{NH}_2)_4]^{2+}$ is less stable as compared to $[\text{Cd(en)}_2]^{2+}$ due to entropy of system is more in case of $[\text{Cd(en)}_2]^{2+}$.
(B) $[\text{Ni(en)}_3]^{2+}$ is less stable than $[\text{Ni}(\text{NH}_3)_6]^{2+}$
(C) $[\text{Co(CN)}_6]^{3-}$ is more stable than $[\text{Rh(CN)}_6]^{3-}$
(D) $[\text{Ag}(\text{NH}_3)_2]^+$ is more stable than $[\text{Ag(en)}]^+$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
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Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

As we know that water is an excellent solvent, water soluble inorganic chemicals that include heavy metals such as cadmium, mercury, nickel etc. constitute an important class of pollutants. All these metals are dangerous to humans because our body cannot excrete them. Over the time, it crosses the tolerance limit. These metals then can damage kidneys, central nervous system, liver etc. Acids (like sulphuric acid) from mine drainage and salts from many different sources including raw salt used to melt snow and ice in the colder climates (sodium and calcium chloride) are water soluble chemical pollutants.

1. The maximum prescribed limit of nitrate ion in drinking water is 'X' mg dm^{-3} and lead is $Y \times 10^{-3} \text{ mg dm}^{-3}$, then calculate (X + Y).
2. The maximum prescribed limit of Fe metal in drinking water is P mg dm^{-3} and Mn is Q mg dm^{-3} . Calculate (P + Q).

Paragraph for Questions 3 and 4

Two complexes (A) and (B) of Cobalt have same molecular formula $\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2\text{ClBr}_2$. 34.15 gm of (A) loses 1.8 gm of weight when heated to 200°C whereas no weight loss was observed on similar treatment with (B).

An aqueous solution of (A) has conductivity equivalent to that of a compound with two ions per formula unit while conductivity of (B) in aqueous solution is equivalent to a compound with (X) number ions per unit. Also a 100 ml 0.1 M aq. solution of (A), on treatment with excess of AgNO_3 . Solution produce (Y) gm of only pale yellow ppt while a 100 ml 0.1 M aqueous solution of (B) on treatment with excess of AgNO_3 solution product 3.315 gm precipitate.

[Molar mass : H = 1, N = 14, O = 16, Cl = 35.5, Co = 59, Br = 80, Ag = 108]

3. the value of (X) is :
4. The value of Y is :

Paragraph for Questions 5 and 6

Different theories like **electrostatic theory** (developed by Kossel, Magnus, Van Arekel, Garrick), **Sidgwick's effective atomic number rule** (EAN), **Pauling's valence bond theory** (VBT), **crystal field theory** (CFT), **adjusted crystal field theory** (ACFT) or **ligand field theory** (LFT) providing some allowance of covalence in CFT, and **molecular orbital theory** (MOT) have been proposed for understanding of bonding in coordination compounds.

5. How many complexes are low spin type ?

$[\text{Co}(\text{NH}_3)_6]^{+3}$, $[\text{CoF}_6]^{-3}$, $[\text{Fe}(\text{CN})_6]^{-4}$, $[\text{Fe}(\text{H}_2\text{O})_6]^{+2}$, $[\text{Cr}(\text{gly})_3]^0$, $[\text{CoBr}_2\text{Cl}_2]^{-2}$

6. Calculate value of (x + y) :

Here

Among the species given below, the total number of diamagnetic species is x.

MnCl_4^{2-} , NO_2 monomer, O_2^- (superoxide), CoCl_4^{2-} , Mn_3O_4 , $(\text{NH}_4)_2[\text{FeCl}_4]$, CuCl_4^{2-} , K_2MnO_4 , K_2CrO_4

The total number of complexes which are following Sidgwick EAN rule is y.

(Atomic number : Mo $\Rightarrow$ 42, Pt $\Rightarrow$ 78, Ag = 47)

$[\text{Mo}(\sigma\text{-C}_3\text{H}_5)\text{Br}(\text{NH}_3)_2]$, $[\text{Fe}(\eta^5\text{-C}_5\text{H}_5)_2]$, $[\text{Ni}(\text{DMG})_2]$, $[\text{Mn}_2(\text{CO})_{10}]$, $[\text{Fe}(\text{CO})_2(\text{NO})_2]^0$, $[\text{PtCl}_3(\eta^2\text{-C}_2\text{H}_4)]^\ominus$, $[\text{Ag}(\text{CN})_2]^{-1}$, $[\text{Fe}_2(\text{CO})_9]^0$

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Calculate the value of $(P \times Q \times R)$

Here

P = Number of green house gases in the following given set of species.

(CO₂, O₂, N₂O, CH₄, CFC, CO, H₂O(g), O₃)

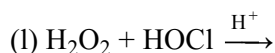
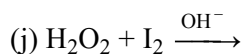
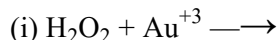
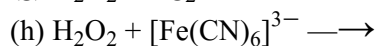
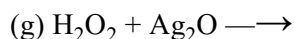
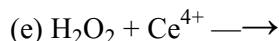
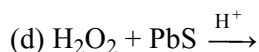
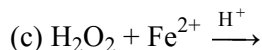
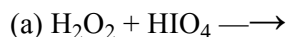
Q = Number of nonviable particulate pollutants in the following given set of species.

(Smoke particles, Fungi, Fly ash, Moulds, Saw dust, Bacteria, Pulverized coke.)

R = Number of Tropospheric gaseous pollutants in the following given set of species.

(SO₂, Mist, CO₂, O₂, NO₂, Smog, CH₄, CO)

8. Consider the following reactions :



If total number of reactions in which H₂O₂ act as reducing agent = X.

If total number of reactions in which H₂O₂ act as oxidizing agent = Y.

Then find the value of X . Y

9. Find the number pair of ions which have same outermost E.C.

Set I : (Ac³⁺, Th⁴⁺), (Th³⁺, Pa⁴⁺), (Pa³⁺, U⁴⁺), (U³⁺, Np⁴⁺), (Np³⁺, Pu⁴⁺), (Pu³⁺, Am⁴⁺), (Am³⁺, Cm⁴⁺), (Cm³⁺, Bk⁴⁺), (Bk³⁺, Cf⁴⁺), (Cf³⁺, Es⁴⁺), (Es³⁺, Fm⁴⁺), (Fm³⁺, Md⁴⁺), (Md³⁺, No⁴⁺), (No³⁺, Lr⁴⁺)

Set II : (La²⁺, Ce³⁺), (Ce²⁺, Pr³⁺), (Pr²⁺, Nd³⁺), (Nd²⁺, Pm³⁺), (Pm²⁺, Sm³⁺), (Sm²⁺, Eu³⁺), (Eu²⁺, Gd³⁺), (Gd²⁺, Tb³⁺), (Tb²⁺, Dy³⁺), (Dy²⁺, Ho³⁺), (Ho²⁺, Er³⁺), (Er²⁺, Tm³⁺), (Tm²⁺, Yb³⁺), (Yb²⁺, Lu³⁺).

If the value of set I is 'x' and set II is 'y', then find |x - y|.

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

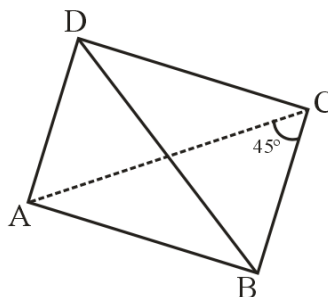
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A drawer contains a mixture of red socks and blue socks, at most 2024 in all. It so happens that, when two socks are selected randomly without replacement, there is a probability of exactly $\frac{1}{2}$ that both are red or both are blue. What is the largest possible number of red socks in the drawer that is consistent with this data ?
(A) 1000 (B) 1012 (C) 990 (D) 946
2. Let $\frac{x-1}{2} = \frac{y-1}{3} = \frac{z+3}{2}$ and $\frac{x-k}{1} = \frac{y+1}{5} = \frac{z}{-1}$ be two lines intersect at (α, β, γ) then $\alpha + \beta + \gamma - k$ equals
(A) 1 (B) 4 (C) 6 (D) 9
3. As shown in the diagram, the volume of tetrahedron DABC is $\frac{1}{6}$. Also $\angle ACB = 45^\circ$ and $AD + BC + \frac{AC}{\sqrt{2}} = 3$, then



- (A) $CD = \sqrt{3}$ (B) $AD = \sqrt{2}$
(C) $BC = 2$ (D) $AC = 2$
4. Position vector of center of circle $|\vec{r}| = 5$ and $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 3\sqrt{3}$ is
(A) $\hat{i} + \hat{j} + \hat{k}$ (B) $3(\hat{i} + \hat{j} + \hat{k})$
(C) $\sqrt{3}(\hat{i} + \hat{j} + \hat{k})$ (D) $\vec{0}$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Let $\vec{a}$ is the only point of intersection of three plane $\vec{r} \cdot \vec{a} = p_1$, $\vec{r} \cdot \vec{b} = p_2$ and $\vec{r} \cdot \vec{c} = p_3$ then which of following is/are correct

(A) $[\vec{a} \ \vec{b} \ \vec{c}] = 0$

(B) $[\vec{a} \ \vec{b} \ \vec{c}] \neq 0$

(C) $\vec{a} = \frac{p_3 (\vec{a} \times \vec{b}) + p_1 (\vec{b} \times \vec{c}) + p_2 (\vec{c} \times \vec{a})}{[\vec{a} \ \vec{b} \ \vec{c}]}$

(D) $\vec{a} = p_3 (\vec{a} \times \vec{b}) + p_1 (\vec{b} \times \vec{c}) + p_2 (\vec{c} \times \vec{a})$

6. Three circles with radii 1, 2 and 3 touch each other externally at point A, B and C, then

(A) Area of $\Delta ABC = 6/5$

(B) Area of $\Delta ABC = 7/5$

(C) Circum-radius of $\Delta ABC = \frac{5}{4}$

(D) Circum-radius of $\Delta ABC = 1$

7. Let A, B and C be three pair-wise independent events such that $P(A \cup B \cup C) = 1$. Then which of the following is/are must be correct ?
- (A) $1 \leq P(A) + P(B) + P(C) \leq 3$
- (B) $P(\bar{A} \cap \bar{B} \cap \bar{C}) = 0$
- (C) $P(A) = P(B) = P(C) = 1$
- (D) $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$
8. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = 2\hat{i} + 3\hat{j} + \hat{k}$, $\vec{c} \times \vec{b} = \vec{d} \times \vec{b}$ and $\vec{d} \cdot \vec{a} = 8$ then which of the following is/are correct ?
- (A) $\vec{d} \cdot \vec{b} = 4$
- (B) $\vec{d} \cdot \vec{c} = 14$
- (C) $[\vec{b} \ \vec{c} \ \vec{d}] = 0$
- (D) $[\vec{a} \ \vec{b} \ \vec{c}] = 2$
9. If A and B are independent events such that the probability that they both occur simultaneously is $\frac{1}{8}$ and the probability that neither of them will occurs is $\frac{3}{8}$, then (A is more probable than B)
- (A) the probability that event A will occur is $\frac{1}{2}$
- (B) the probability that event B will occur is $\frac{1}{4}$
- (C) the probability that event A will occur is less than $\frac{3}{4}$
- (D) the probability that event B will occur is more than $\frac{1}{3}$
10. Suppose you play a game by tossing a fair coin repeatedly and independently. If a head comes up, you win Rs.1 and if a tail comes up you lose Rs.1. Suppose you start with Rs. n. Let $P(m, n)$ be the probability that you will get to Rs. m first without running out of money ($m, n \in \mathbb{N}, m > n$). Which of the following is/are CORRECT ?
- (A) $P(51, 50) = \frac{1}{2}$
- (B) $P(325, 52) = \frac{4}{25}$
- (C) $P(100, 50) \geq \frac{1}{2}$
- (D) $P(325, 52) = \frac{7}{25}$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

In $\triangle ABC$, $AB = AC$. Point D and E lie on BC such that $BD = DC$ and $BE > CE$. $\tan \angle EAC$, $\tan \angle EAD$ and $\tan \angle EAB$ are in G.P.

1. $\angle EAD$ is equal to $\frac{\pi}{k}$, then the value of k is
2. If $AE = 10$ then AD equals $\lambda\sqrt{2}$, then the value of λ is

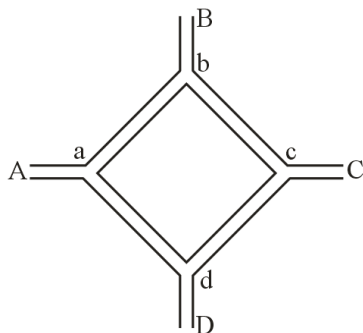
Paragraph for Questions 3 and 4

Let a line with direction ratio 2,2,1 intersect the line $\frac{x-7}{3} = \frac{y-5}{2} = \frac{z-3}{1}$ and $\frac{x-1}{2} = \frac{y+1}{4} = \frac{z+1}{3}$ at A and B respectively

3. If (x_1, y_1, z_1) is A's coordinate. Then find $x_1 + y_1 + z_1 =$
4. Distance AB is equal to

Paragraph for Questions 5 and 6

Alice is in a wonderland of tunnels show below. At each intersection (a,b,c,d) he chooses a direction randomly (including possibly turning around). Once Alice reaches an exit, she will not return into the tunnels.



Let $P(x_y)$ = Probability of exiting at Y when currently at the intersection x.

Find

5. $15P(A_a)$ is equal to
6. $15P(b_A)$ is equal to

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. Let A and B be points (2,0,0) and (0,4,0) respectively. If E is mid-point of AB and F is mirror image of origin with respect edge AB and let D the point on the line $\frac{x-1}{0} = \frac{y-2}{0} = \frac{z}{1}$ at 6 unit distance from the edge AB, then volume of tetrahedron DFAB is _____
8. Let O-ABC be a regular tetrahedron (A ($\vec{a}$), B ($\vec{b}$), C ($\vec{c}$). O is origin) with edge length 1. Find $4|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|^2 = ?$
9. From a bag A containing 3 white and 5 black balls, 4 balls are transferred into an empty bag B. From bag B 2 balls are taken and they both happen to be white. These balls are returned to the bag B. Now a ball is taken from bag B is white; the probability of this event can be expressed as p/q where p and q are relatively prime positive integers. Find (q - p).

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work

ALLEN CAREER INSTITUTE Pvt. Ltd.

Registered & Corporate Office : 'SANKALP', CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005

Ph. : +91-744-3556677, +91-744-2757575 | E-mail : info@allen.in | Website : www.allen.ac.in

E-24/24

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